

2 - Domain Specific IoTs

This Chapter Covers

IoT Applicatins for:

- Home
- Cities
- Environment
- Energy Systems
- Retail
- Logistics
- Industry
- Agriculture
- Health & Lifestyle

2.1 Introduction

The Internet of Things (IoT) applications span a wide range of domains including (but not limited to) homes, cities, environment, energy systems, retail, logistics, industry, agriculture and health. This chapter provides an overview of various types of IoT applications for each of these domains. In the later chapters the reader is guided through detailed implementations of several of these applications.

2.2 Home Automation

2.2.1 Smart Lighting

Smart lighting for homes helps in saving energy by adapting the lighting to the ambient conditions and switching on/off or dimming the lights when needed. Key enabling technologies for smart lighting include solid state lighting (such as LED lights) and IP-enabled lights. For solid state lighting solutions both spectral and temporal characteristics can be configured to adapt illumination to various needs. Smart lighting solutions for home achieve energy savings by sensing the human movements and their environments and controlling the lights accordingly. Wireless-enabled and Internet connected lights can be controlled remotely from IoT applications such as a mobile or web application. Smart lights with sensors for occupancy, temperature, lux level, etc., can be configured to adapt the lighting (by changing the light intensity, color, etc.) based on the ambient conditions sensed, in order to provide a good ambiance. In [19] controllable LED lighting system is presented that is embedded with ambient intelligence gathered from a distributed smart wireless sensor network to optimize and control the lighting system to be more efficient and user-oriented. A solid state lighting model is described in [20] and implemented on a wireless sensor network that provides services for sensing illumination changes and dynamically adjusting luminary brightness according to user preferences. In chapter-9 we provide a case study on a smart lighting system.

2.2.2 Smart Appliances

Modern homes have a number of appliances such as TVs, refrigerators, music systems, washer/dryers, etc. Managing and controlling these appliances can be cumbersome, with each appliance having its own controls or remote controls. Smart appliances make the management easier and also provide status information to the users remotely. For example, smart washer/dryers that can be controlled remotely and notify when the washing/drying cycle is complete. Smart thermostats allow controlling the temperature remotely and can learn the user preferences [22]. Smart refrigerators can keep track of the items stored (using RFID tags) and send updates to the users when an item is low on stock. Smart

TVs allows users to search and stream videos and movies from the Internet on a local storage drive, search TV channel schedules and fetch news, weather updates and other content from the Internet. OpenRemote [21] is an open source automation platform for homes and buildings. OpenRemote is platform agnostic and works with standard hardware. With OpenRemote, users can control various appliances using mobile or web applications. OpenRemote comprises of three components - a Controller that manages scheduling and runtime integration between devices, a Designer that allows you to create both configurations for the controller and create user interface designs and Control Panels that allow you to interact with devices and control them. An IoT-based appliance control system for smart homes is described in [23], that uses a smart central controller to set up a wireless sensor and actuator network and control modules for appliances.

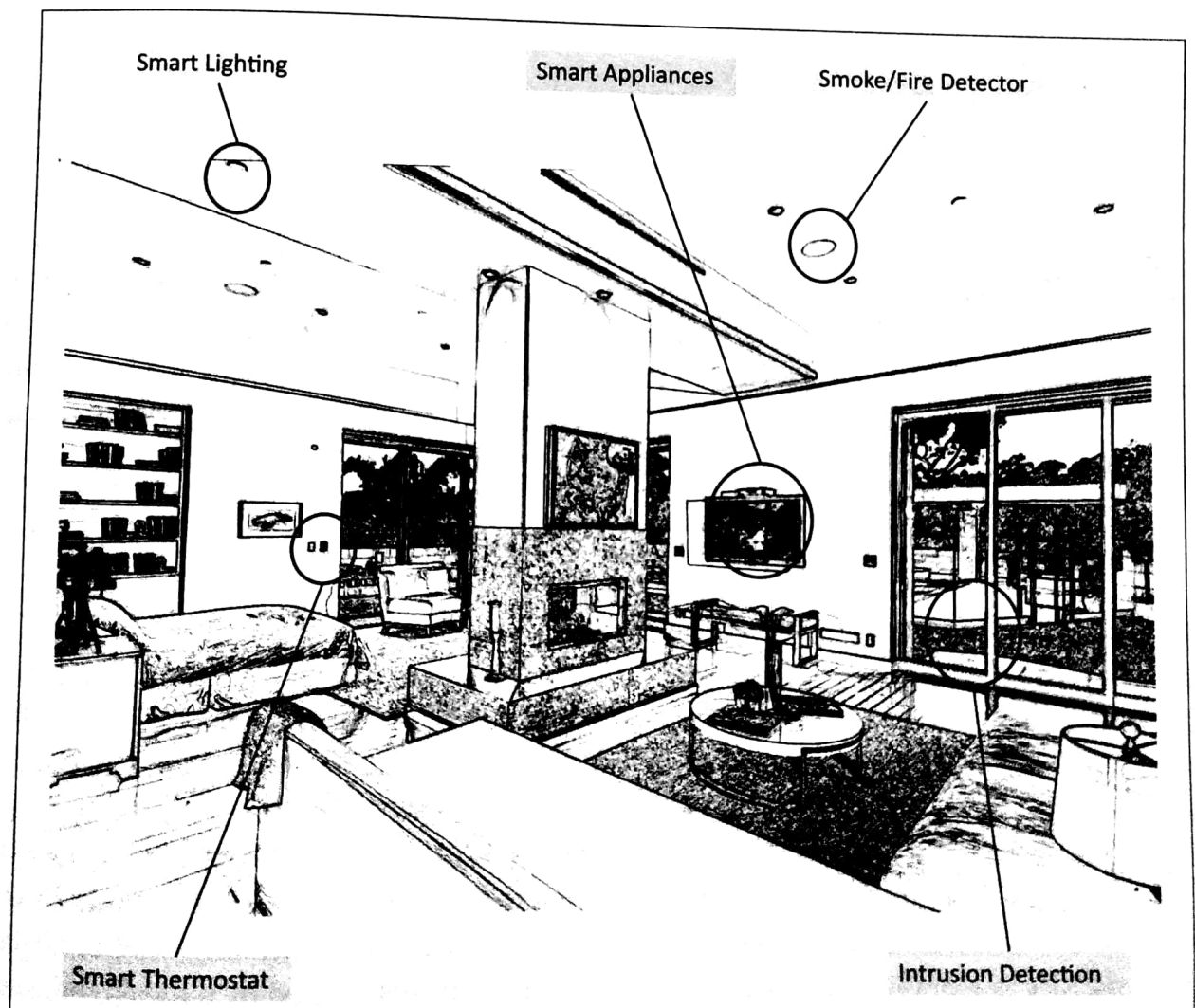


Figure 2.1: Applications of IoT for homes

2.2.3 Intrusion Detection

Home intrusion detection systems use security cameras and sensors (such as PIR sensors and door sensors) to detect intrusions and raise alerts. Alerts can be in the form of an SMS or an email sent to the user. Advanced systems can even send detailed alerts such as an image grab or a short video clip sent as an email attachment. A cloud controlled intrusion detection system is described in [24] that uses location-aware services, where the geo-location of each node of a home automation system is independently detected and stored in the cloud. In the event of intrusions, the cloud services alert the accurate neighbors (who are using the home automation system) or local police. In [25], an intrusion detection system based on UPnP technology is described. The system uses image processing to recognize the intrusion and extract the intrusion subject and generate Universal-Plug-and-Play (UPnP-based) instant messaging for alerts. In chapter-9 we provide a case study on an intrusion detection system.

2.2.4 Smoke/Gas Detectors

Smoke detectors are installed in homes and buildings to detect smoke that is typically an early sign of fire. Smoke detectors use optical detection, ionization or air sampling techniques to detect smoke. Alerts raised by smoke detectors can be in the form of signals to a fire alarm system. Gas detectors can detect the presence of harmful gases such as carbon monoxide (CO), liquid petroleum gas (LPG), etc. A smart smoke/gas detector [22] can raise alerts in human voice describing where the problem is, send or an SMS or email to the user or the local fire safety department and provide visual feedback on its status (healthy, battery-low, etc.). In [26], the design of a system that detects gas leakage and smoke and gives visual level indication, is described.

2.3 Cities

2.3.1 Smart Parking

Finding a parking space during rush hours in crowded cities can be time consuming and frustrating. Furthermore, drivers blindly searching for parking spaces create additional traffic congestion. Smart parking make the search for parking space easier and convenient for drivers. Smart parking are powered by IoT systems that detect the number of empty parking slots and send the information over the Internet to smart parking application back-ends. These applications can be accessed by the drivers from smart-phones, tablets and in-car navigation systems. In smart parking, sensors are used for each parking slot, to detect whether the slot is empty or occupied. This information is aggregated by a local controller and then sent over the Internet to the database. In [29], Polycarpou *et. al.* describe latest trends in parking availability monitoring, parking reservation and dynamic pricing schemes.

Design and implementation of a prototype smart parking system based on wireless sensor network technology with features like remote parking monitoring, automated guidance, and parking reservation mechanism is described in [30]. In chapter-9 we provide a case study on a smart parking system.

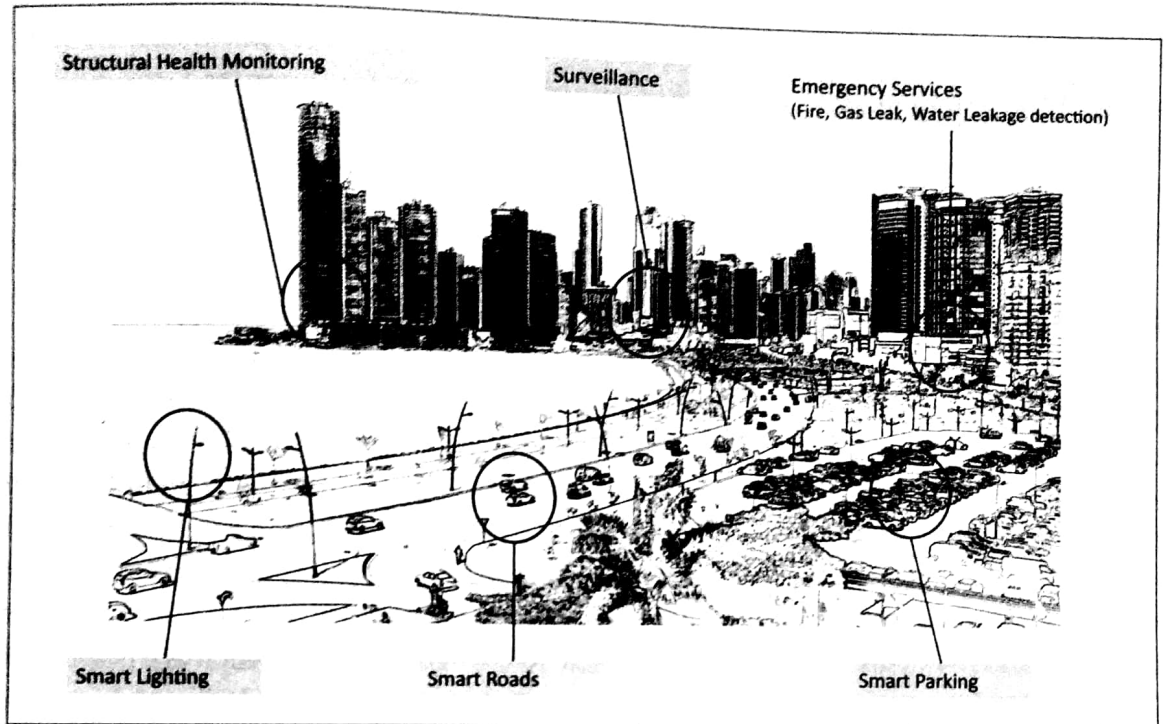


Figure 2.2: Applications of IoT for cities

2.3.2 Smart Lighting

Smart lighting systems for roads, parks and buildings can help in saving energy. According to an IEA report [27], lighting is responsible for 19% of global electricity use and around 6% of global greenhouse gas emissions. Smart lighting allows lighting to be dynamically controlled and also adaptive to the ambient conditions. Smart lights connected to the Internet can be controlled remotely to configure lighting schedules and lighting intensity. Custom lighting configurations can be set for different situations such as a foggy day, a festival, etc. Smart lights equipped with sensors can communicate with other lights and exchange information on the sensed ambient conditions to adapt the lighting. Castro *et. al.* [28] describe the need for smart lighting system in smart cities, smart lighting features and how to develop interoperable smart lighting solutions.

2.3.3 Smart Roads

Smart roads equipped with sensors can provide information on driving conditions, travel time estimates and alerts in case of poor driving conditions, traffic congestions and accidents. Such information can help in making the roads safer and help in reducing traffic jams. Information sensed from the roads can be communicated via Internet to cloud-based applications and social media and disseminated to the drivers who subscribe to such applications. In [31], a distributed and autonomous system of sensor network nodes for improving driving safety on public roads is proposed. The system can provide the drivers and passengers with a consistent view of the road situation a few hundred meters ahead of them or a few dozen miles away, so that they can react to potential dangers early enough.

2.3.4 Structural Health Monitoring

Structural Health Monitoring systems use a network of sensors to monitor the vibration levels in the structures such as bridges and buildings. The data collected from these sensors is analyzed to assess the health of the structures. By analyzing the data it is possible to detect cracks and mechanical breakdowns, locate the damages to a structure and also calculate the remaining life of the structure. Using such systems, advance warnings can be given in the case of imminent failure of the structure. An environmental effect removal based structural health monitoring scheme in an IoT environment is proposed in [32]. Since structural health monitoring systems use large number of wireless sensor nodes which are powered by traditional batteries, researchers are exploring energy harvesting technologies to harvesting ambient energy, such as mechanical vibrations, sunlight, and wind [33, 34].

2.3.5 Surveillance

Surveillance of infrastructure, public transport and events in cities is required to ensure safety and security. City wide surveillance infrastructure comprising of large number of distributed and Internet connected video surveillance cameras can be created. The video feeds from surveillance cameras can be aggregated in cloud-based scalable storage solutions. Cloud-based video analytics applications can be developed to search for patterns or specific events from the video feeds. In [35] a smart city surveillance system is described that leverages benefits of cloud data stores.

2.3.6 Emergency Response

IoT systems can be used for monitoring the critical infrastructure in cities such as buildings, gas and water pipelines, public transport and power substations. IoT systems for fire detection, gas and water leakage detection can help in generating alerts and minimizing

their effects on the critical infrastructure. IoT systems for critical infrastructure monitoring enable aggregation and sharing of information collected from large number of sensors. Using cloud-based architectures, multi-modal information such as sensor data, audio, video feeds can be analyzed in near real-time to detect adverse events. Response to alerts generated by such systems can be in the form of alerts sent to the public, re-routing of traffic, evacuations of the affected areas, etc. In [36] Attwood *et. al.* describe critical infrastructure response framework for smart cities. A Traffic Management System for emergency services is described in [37]. The system adapts by dynamically adjusting traffic lights, changing related driving policies, recommending behavior change to drivers, and applying essential security controls. Such systems can reduce the latency of emergency services for vehicles such as ambulances and police cars while minimizing disruption of regular traffic.

2.4 Environment

2.4.1 Weather Monitoring

IoT-based weather monitoring systems can collect data from a number of sensor attached (such as temperature, humidity, pressure, etc.) and send the data to cloud-based applications and storage back-ends. The data collected in the cloud can then be analyzed and visualized by cloud-based applications. Weather alerts can be sent to the subscribed users from such applications. AirPi [38] is a weather and air quality monitoring kit capable of recording and uploading information about temperature, humidity, air pressure, light levels, UV levels, carbon monoxide, nitrogen dioxide and smoke level to the Internet. In [39], a pervasive weather monitoring system is described that is integrated with buses to measure weather variables like humidity, temperature and air quality during the bus path. In [40], a weather monitoring system based on wireless sensor networks is described. In chapter-9 we provide a case study on a weather monitoring system.

2.4.2 Air Pollution Monitoring

IoT based air pollution monitoring systems can monitor emission of harmful gases (CO_2 , CO , NO , NO_2 , etc.) by factories and automobiles using gaseous and meteorological sensors. The collected data can be analyzed to make informed decisions on pollutions control approaches. In [41], a real-time air quality monitoring system is presented that comprises of several distributed monitoring stations that communicate via wireless with a back-end server using machine-to-machine communication. In [42], an air pollution system is described that integrates a single-chip microcontroller, several air pollution sensors, GPRS-Modem, and a GPS module. In chapter-9 we provide a case study on an air pollution monitoring system.

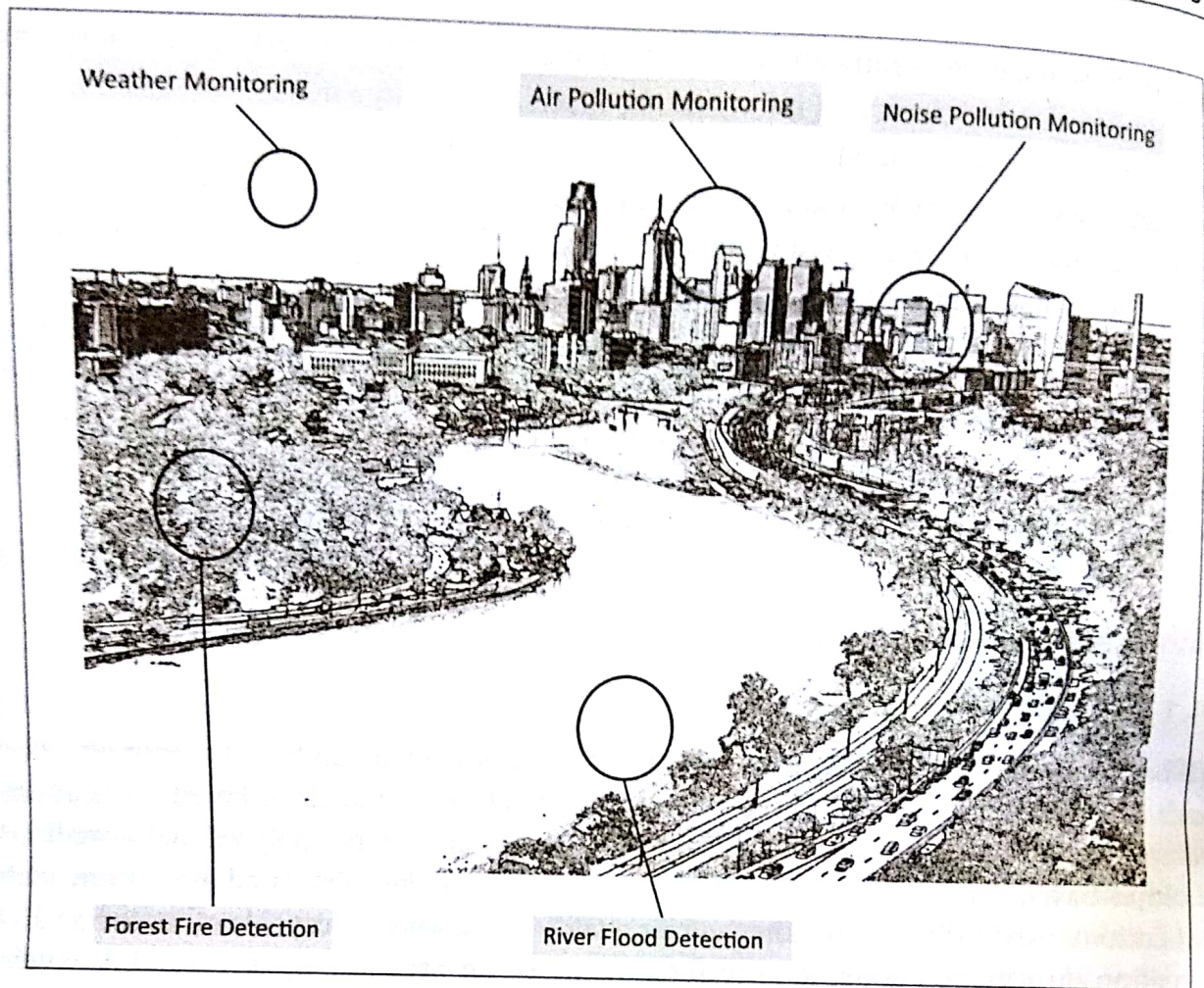


Figure 2.3: Applications of IoT for environment

2.4.3 Noise Pollution Monitoring

Due to growing urban development, noise levels in cities have increased and even become alarmingly high in some cities. Noise pollution can cause health hazards for humans due to sleep disruption and stress. Noise pollution monitoring can help in generating noise maps for cities. Urban noise maps can help the policy makers in urban planning and making policies to control noise levels near residential areas, schools and parks. IoT based noise pollution monitoring systems use a number of noise monitoring stations that are deployed at different places in a city. The data on noise levels from the stations is collected on servers or in the cloud. The collected data is then aggregated to generate noise maps. In [43], a noise mapping study for a city is presented which revealed that the city suffered from serious noise pollution. In [44], the design of smart phone application is described that allows the users to continuously measure noise levels and send to a central server where all generated

information is aggregated and mapped to a meaningful noise visualization map.

2.4.4 Forest Fire Detection

Forest fires can cause damage to natural resources, property and human life. There can be different causes of forest fires including lightening, human negligence, volcanic eruptions and sparks from rock falls. Early detection of forest fires can help in minimizing the damage. IoT based forest fire detection systems use a number of monitoring nodes deployed at different locations in a forest. Each monitoring node collects measurements on ambient conditions including temperature, humidity, light levels, etc. A system for early detection of forest fires is described in [45] that provides early warning of a potential forest fire and estimates the scale and intensity of the fire if it materializes. In [46], a forest fire detection system based on wireless sensor networks is presented. The system uses multi-criteria detection which is implemented by the artificial neural network (ANN). The ANN fuses sensing data corresponding to multiple attributes of a forest fire (such as temperature, humidity, infrared and visible light) to detect forest fires.

2.4.5 River Floods Detection

River floods can cause extensive damage to the natural and human resources and human life. River floods occur due to continuous rainfall which cause the river levels to rise and flow rates to increase rapidly. Early warnings of floods can be given by monitoring the water level and flow rate. IoT based river flood monitoring system use a number of sensor nodes that monitor the water level (using ultrasonic sensors) and flow rate (using the flow velocity sensors). Data from a number of such sensor nodes is aggregated in a server or in the cloud. Monitoring applications raise alerts when rapid increase in water level and flow rate is detected. In [47], a river flood monitoring system is described that measures river and weather conditions through wireless sensor nodes equipped with different sensors. In [48], a motes-based sensor network for river flood monitoring is described. The system includes a water level monitoring module, network video recorder module, and data processing module that provides flood information in the form of raw data, predicted data, and video feed.

2.5 Energy

2.5.1 Smart Grids

Smart Grid is a data communications network integrated with the electrical grid that collects and analyzes data captured in near-real-time about power transmission, distribution, and consumption. Smart Grid technology provides predictive information and recommendations to utilities, their suppliers, and their customers on how best to manage power. Smart

Grids collect data regarding electricity generation (centralized or distributed), consumption (instantaneous or predictive), storage (or conversion of energy into other forms), distribution and equipment health data. Smart grids use high-speed, fully integrated, two-way communication technologies for real-time information and power exchange. By using IoT based sensing and measurement technologies, the health of equipment and the integrity of the grid can be evaluated. Smart meters can capture almost real-time consumption, remotely control the consumption of electricity and remotely switch off supply when required. Power thefts can be prevented using smart metering. By analyzing the data on power generation, transmission and consumption smart grids can improve efficiency throughout the electric system. Storage collection and analysis of smart grids data in the cloud can help in dynamic optimization of system operations, maintenance, and planning. Cloud-based monitoring of smart grids data can improve energy usage levels via energy feedback to users coupled with real-time pricing information. Real-time demand response and management strategies can be used for lowering peak demand and overall load via appliance control and energy storage mechanisms. Condition monitoring data collected from power generation and transmission systems can help in detecting faults and predicting outages. In [49], application of IoT in smart grid power transmission is described.

2.5.2 Renewable Energy Systems

Due to the variability in the output from renewable energy sources (such as solar and wind), integrating them into the grid can cause grid stability and reliability problems. Variable output produces local voltage swings that can impact power quality. Existing grids were designed to handle power flows from centralized generation sources to the loads through transmission and distribution lines. When distributed renewable energy sources are integrated into the grid, they create power bi-directional power flows for which the grids were not originally designed. IoT based systems integrated with the transformers at the point of interconnection measure the electrical variables and how much power is fed into the grid. To ensure the grid stability, one solution is to simply cut off the overproduction. For wind energy systems, closed-loop controls can be used to regulate the voltage at point of interconnection which coordinate wind turbine outputs and provides reactive power support [52].

2.5.3 Prognostics

Energy systems (smart grids, power plants, wind turbine farms, for instance) have a large number of critical components that must function correctly so that the systems can perform their operations correctly. For example, a wind turbine has a number of critical components, e.g., bearings, turning gears, for instance, that must be monitored carefully as wear and tear in such critical components or sudden change in operating conditions of the machines can

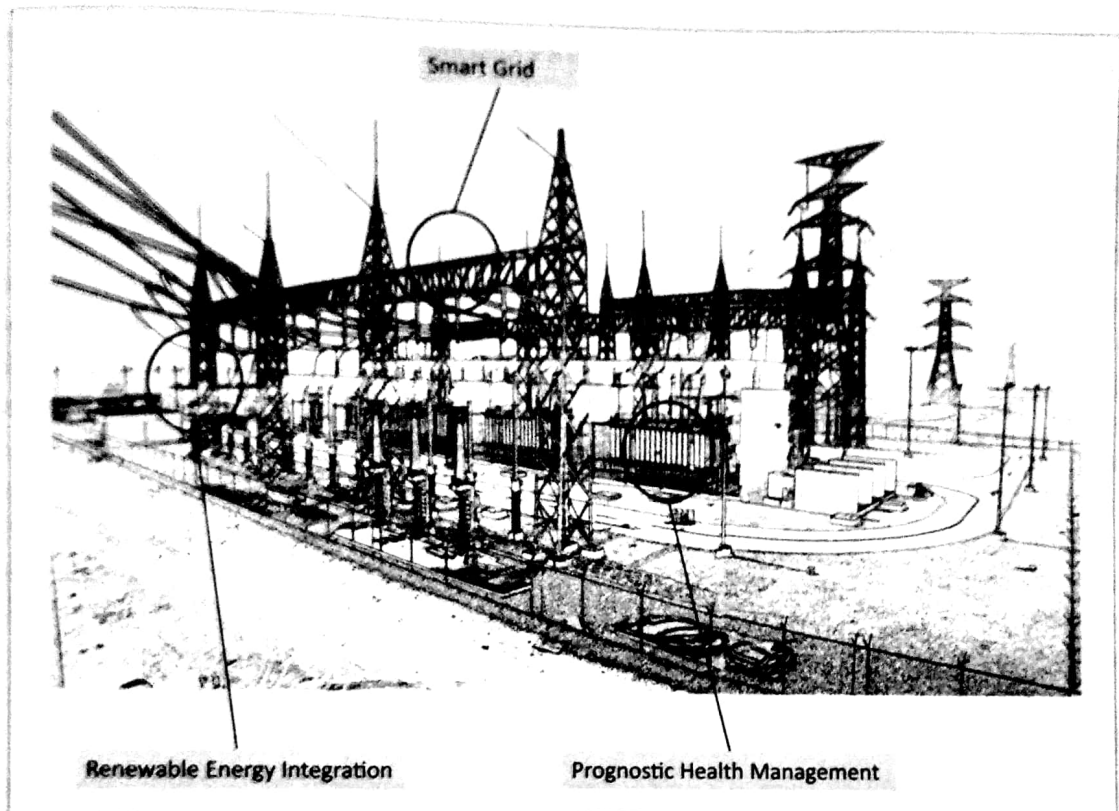


Figure 2.4: Applications of IoT for energy systems

result in failures. In systems such as power grids, real-time information is collected using specialized electrical sensors called Phasor Measurement Units (PMU) at the substations. The information received from PMUs must be monitored in real-time for estimating the state of the system and for predicting failures. Energy systems have thousands of sensors that gather real-time maintenance data continuously for condition monitoring and failure prediction purposes. IoT based prognostic real-time health management systems can predict performance of machines or energy systems by analyzing the extent of deviation of a system from its normal operating profiles. Analyzing massive amounts of maintenance data collected from sensors in energy systems and equipment can provide predictions for the impending failures (potentially in real-time) so that their reliability and availability can be improved. Prognostic health management systems have been developed for different energy systems. OpenPDC [50] is a set of applications for processing of streaming time-series data collected from Phasor Measurement Units (PMUs) in real-time. A generic framework for storage, processing and analysis of massive machine maintenance data, collected from a large number

of sensors embedded in industrial machines, in a cloud computing environment was proposed in [51].

2.6 Retail

2.6.1 Inventory Management

Inventory management for retail has become increasingly important in the recent years with the growing competition. While over-stocking of products can result in additional storage expenses and risk (in case of perishables), under-stocking can lead to loss of revenue. IoT systems using Radio Frequency Identification (RFID) tags can help in inventory management and maintaining the right inventory levels. RFID tags attached to the products allow them to be tracked in real-time so that the inventory levels can be determined accurately and products which are low on stock can be replenished. Tracking can be done using RFID readers attached to the retail store shelves or in the warehouse. IoT systems enable remote monitoring of inventory using the data collected by the RFID readers. In [53], an RFID data-based inventory management system for time-sensitive materials is described.

2.6.2 Smart Payments

Smart payment solutions such as contact-less payments powered by technologies such as Near field communication (NFC) and Bluetooth. Near field communication (NFC) is a set of standards for smart-phones and other devices to communicate with each other by bringing them into proximity or by touching them. Customers can store the credit card information in their NFC-enabled smart-phones and make payments by bringing the smart-phones near the point of sale terminals. NFC maybe used in combination with Bluetooth, where NFC (which offers low speeds) initiates initial pairing of devices to establish a Bluetooth connection while the actual data transfer takes place over Bluetooth. The applications of NFC for contact-less payments are described in [54, 55].

2.6.3 Smart Vending Machines

Smart vending machines connected to the Internet allow remote monitoring of inventory levels, elastic pricing of products, promotions, and contact-less payments using NFC. Smart-phone applications that communicate with smart vending machines allow user preferences to be remembered and learned with time. When a user moves from one vending machine to the other and pairs the smart-phone with the vending machine, a user specific interface is presented. Users can save their preferences and favorite products. Sensors in a smart vending machine monitor its operations and send the data to the cloud which can be used for predictive maintenance. Smart vending machines can communicate with other

vending machines in their vicinity and share their inventory levels so that the customers can be routed to the nearest machine in case a product goes out of stock in a machine. For perishable items, the smart vending machines can reduce the price as the expiry date nears. New products can be recommended to the customers based on the purchase history and preferences.

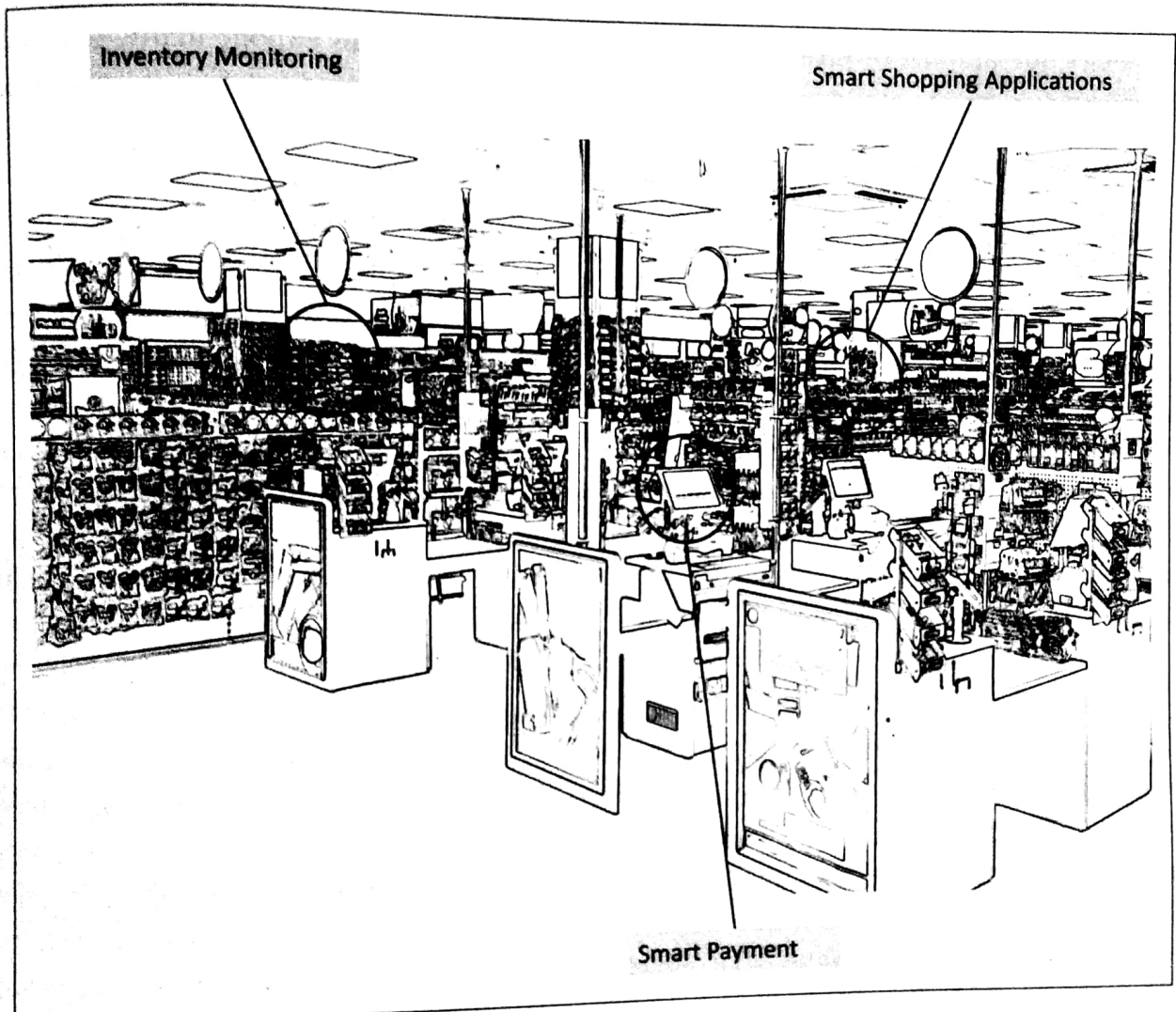


Figure 2.5: Applications of IoT for retail

2.7 Logistics

2.7.1 Route Generation & Scheduling

Modern transportation systems are driven by data collected from multiple sources which is processed to provide new services to the stakeholders. By collecting large amount of

data from various sources and processing the data into useful information, data-driven transportation systems can provide new services such as advanced route guidance [62, 63], dynamic vehicle routing [64], anticipating customer demands for pickup and delivery problem, for instance. Route generation and scheduling systems can generate end-to-end routes using combination of route patterns and transportation modes and feasible schedules based on the availability of vehicles. As the transportation network grows in size and complexity, the number of possible route combinations increases exponentially. IoT based systems backed by the cloud can provide fast response to the route generation queries and can be scaled up to serve a large transportation network.

2.7.2 Fleet Tracking

Vehicle fleet tracking systems use GPS technology to track the locations of the vehicles in real-time. Cloud-based fleet tracking systems can be scaled up on demand to handle large number of vehicles. Alerts can be generated in case of deviations in planned routes. The vehicle locations and routes data can be aggregated and analyzed for detecting bottlenecks in the supply chain such as traffic congestions on routes, assignments and generation of alternative routes, and supply chain optimization. In [58], a fleet tracking system for commercial vehicles is described. The system can analyze messages sent from the vehicles to identify unexpected incidents and discrepancies between actual and planned data, so that remedial actions can be taken.

2.7.3 Shipment Monitoring

Shipment monitoring solutions for transportation systems allow monitoring the conditions inside containers. For example, containers carrying fresh food produce can be monitored to prevent spoilage of food. IoT based shipment monitoring systems use sensors such as temperature, pressure, humidity, for instance, to monitor the conditions inside the containers and send the data to the cloud, where it can be analyzed to detect food spoilage. The analysis and interpretation of data on the environmental conditions in the container and food truck positioning can enable more effective routing decisions in real time. Therefore, it is possible to take remedial measures such as - the food that has a limited time budget before it gets rotten can be re-routed to a closer destinations, alerts can be raised to the driver and the distributor about the transit conditions, such as container temperature exceeding the allowed limit, humidity levels going out of the allowed limit, for instance, and corrective actions can be taken before the food gets damaged. A cloud-based framework for real-time fresh food supply tracking and monitoring was proposed in [61]. For fragile products, vibration levels during shipments can be tracked using accelerometer and gyroscope sensors attached to IoT devices. In [59], a system for monitoring container integrity and operating conditions

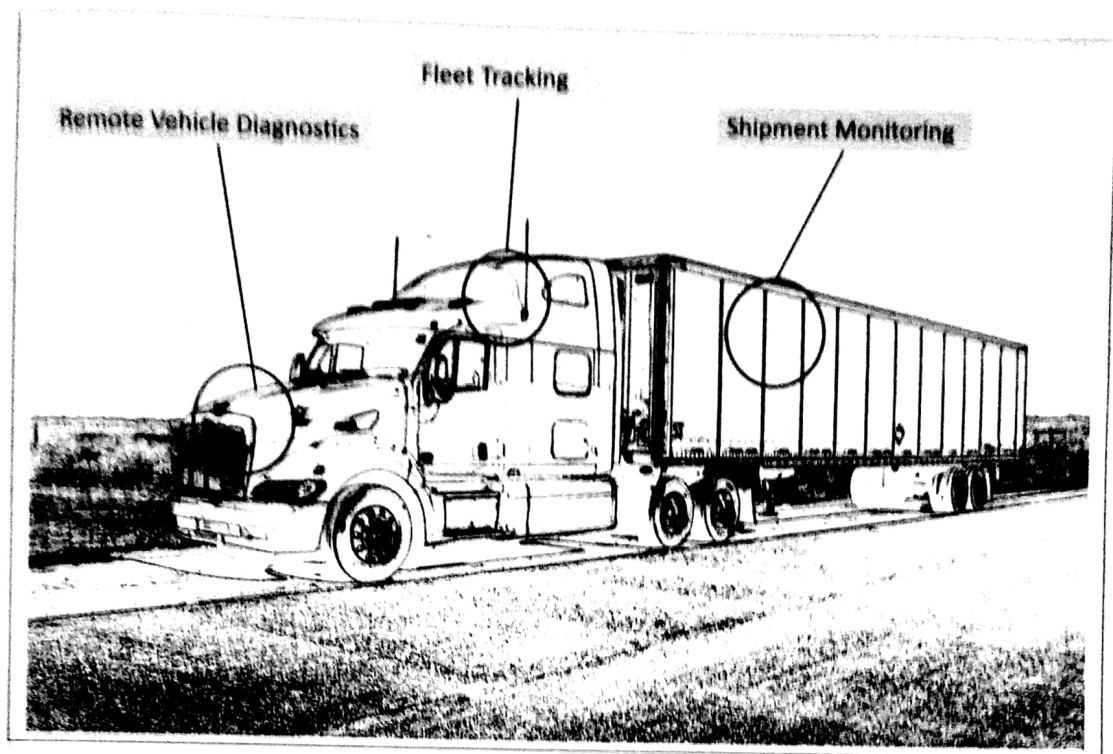


Figure 2.6: Applications of IoT for logistics

is described. The system monitors the vibration patterns of a container and its contents to reveal information related to its operating environment and integrity during transport, handling and storage.

2.7.4 Remote Vehicle Diagnostics

Remote vehicle diagnostic systems can detect faults in the vehicles or warn of impending faults. These diagnostic systems use on-board IoT devices for collecting data on vehicle operation (such as speed, engine RPM, coolant temperature, fault code number) and status of various vehicle sub-systems. Such data can be captured by integrating on-board diagnostic systems with IoT devices using protocols such as CAN bus. Modern commercial vehicles support on-board diagnostic (OBD) standards such as OBD-II. OBD systems provide real-time data on the status of vehicle sub-systems and diagnostic trouble codes which allow rapidly identifying the faults in the vehicle. IoT based vehicle diagnostic systems can send the vehicle data to centralized servers or the cloud where it can be analyzed to generate alerts and suggest remedial actions. In [60], a real-time online vehicle diagnostics and early fault estimation system is described. The system makes use of on-board vehicle diagnostics

device and expert system to achieve real-time vehicle diagnostics and fault warning.

2.8 Agriculture

2.8.1 Smart Irrigation

Smart irrigation systems can improve crop yields while saving water. Smart irrigation systems use IoT devices with soil moisture sensors to determine the amount of moisture in the soil and release the flow of water through the irrigation pipes only when the moisture levels go below a predefined threshold. Smart irrigation systems also collect moisture level measurements on a server or in the cloud where the collected data can be analyzed to plan watering schedules. Cultivar's RainCloud [56] is a device for smart irrigation that uses water valves, soil sensors and a WiFi enabled programmable computer.

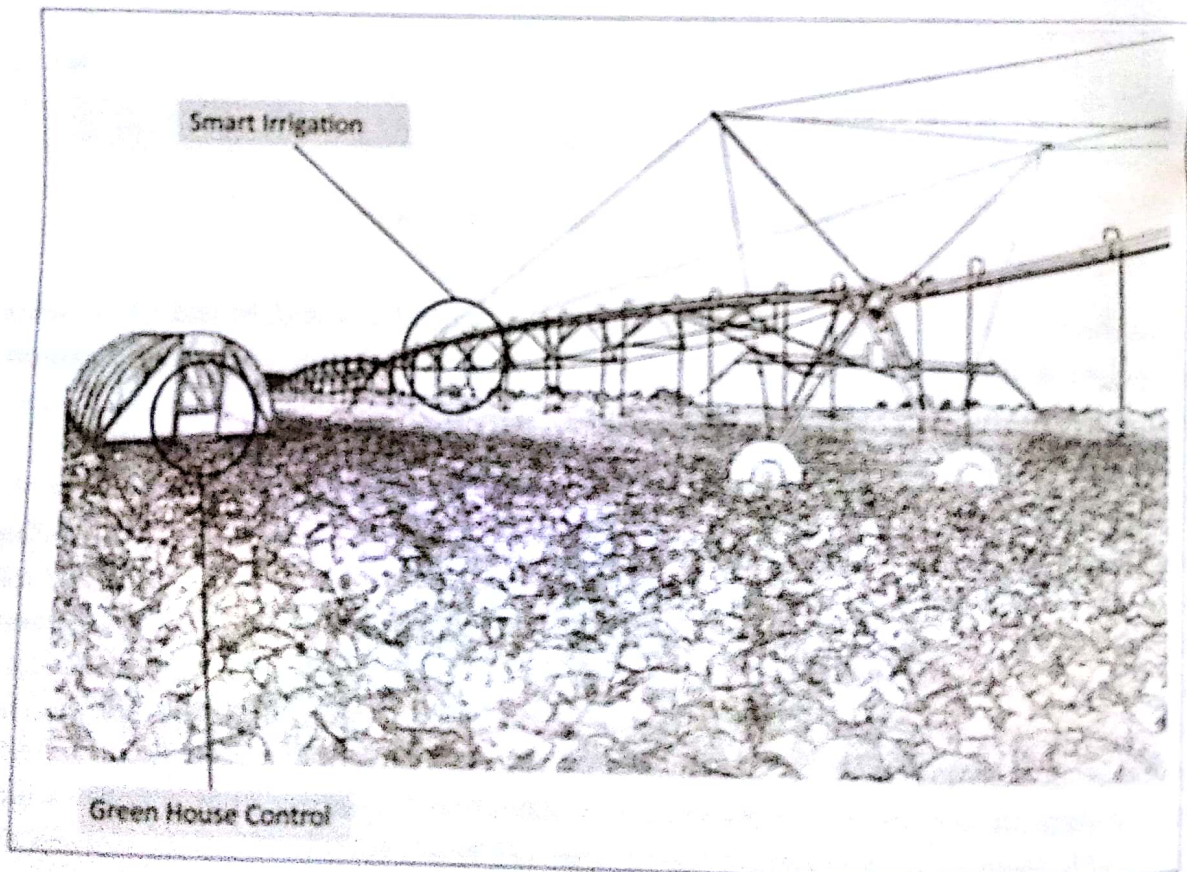


Figure 2.7: Applications of IoT for agriculture

2.8.2 Green House Control

Green houses are structures with glass or plastic roofs that provide conducive environment for growth of plants. The climatological conditions inside a green house can be monitored and controlled to provide the best conditions for growth of plants. The temperature, humidity, soil moisture, light and carbon dioxide levels are monitored using sensors and the climatological conditions are controlled automatically using actuation devices (such as valves for releasing water and switches for controlling fans). IoT systems play an important role in green house control and help in improving productivity. The data collected from various sensors is stored on centralized servers or in the cloud where analysis is performed to optimize the control strategies and also correlate the productivity with different control strategies. In [57], the design of a wireless sensing and control system for precision green house management is described. The system uses wireless sensor network to monitor and control the agricultural parameters like temperature and humidity in real time for better management and maintenance of agricultural production.

2.9 Industry

2.9.1 Machine Diagnosis & Prognosis

Machine prognosis refers to predicting the performance of a machine by analyzing the data on the current operating conditions and how much deviations exist from the normal operating conditions. Machine diagnosis refers to determining the cause of a machine fault. IoT plays a major role in both prognosis and diagnosis of industrial machines. Industrial machines have a large number of components that must function correctly for the machine to perform its operations. Sensors in machines can monitor the operating conditions such as (temperature and vibration levels). The sensor data measurements are done on timescales of few milliseconds to few seconds, which leads to generation of massive amount of data. IoT based systems integrated with cloud-based storage and analytics back-ends can help in storage, collection and analysis of such massive scale machine sensor data. A number of methods have been proposed for reliability analysis and fault prediction in machines. Case-based reasoning (CBR) is a commonly used method that finds solutions to new problems based on past experience. This past experience is organized and represented as cases in a case-base. CBR is an effective technique for problem solving in the fields in which it is hard to establish a quantitative mathematical model, such as machine diagnosis and prognosis. Since for each machine, data from a very large number of sensors is collected, using such high dimensional data for creation of case library reduces the case retrieval efficiency. Therefore, data reduction and feature extraction methods are used to find the representative set of features which have the same classification ability as the complete of

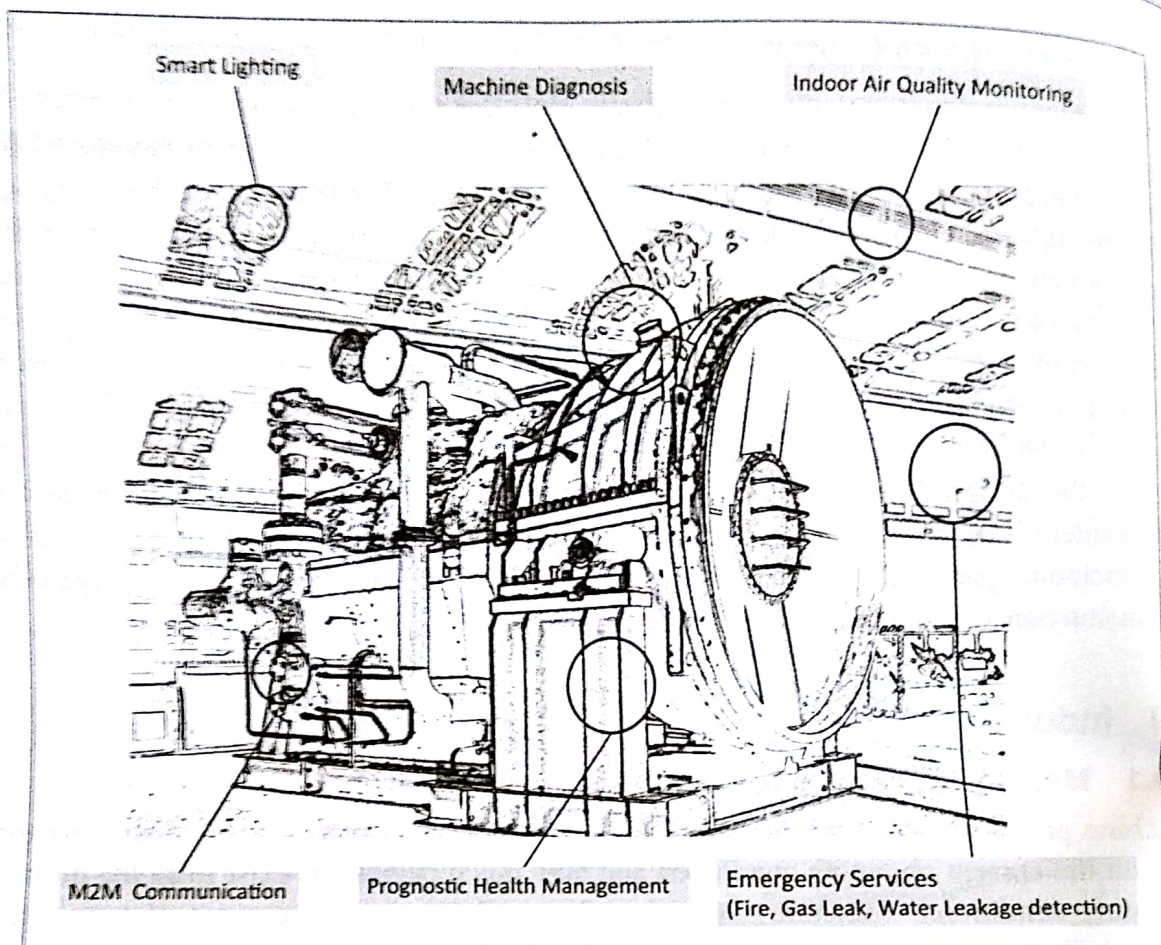


Figure 2.8: Applications of IoT for industry

features. A CBR based machine fault diagnosis and prognosis approach is described in [51]. A survey on recent trends in machine diagnosis and prognosis algorithms is presented in [65].

2.9.2 Indoor Air Quality Monitoring

Monitoring indoor air quality in factories is important for health and safety of the workers. Harmful and toxic gases such as carbon monoxide (CO), nitrogen monoxide (NO), Nitrogen Dioxide (NO_2), etc., can cause serious health problems. IoT based gas monitoring systems can help in monitoring the indoor air quality using various gas sensors. The indoor air quality can vary for different locations. Wireless sensor networks based IoT devices can identify the hazardous zones, so that corrective measures can be taken to ensure proper ventilation. In [66] a hybrid sensor system for indoor air quality monitoring is presented, which contains both stationary sensors (for accurate readings and calibration) and mobile

sensors (for coverage). In [67] a wireless solution for indoor air quality monitoring is described that measures the environmental parameters like temperature, humidity, gaseous pollutants, aerosol and particulate matter to determine the indoor air quality.

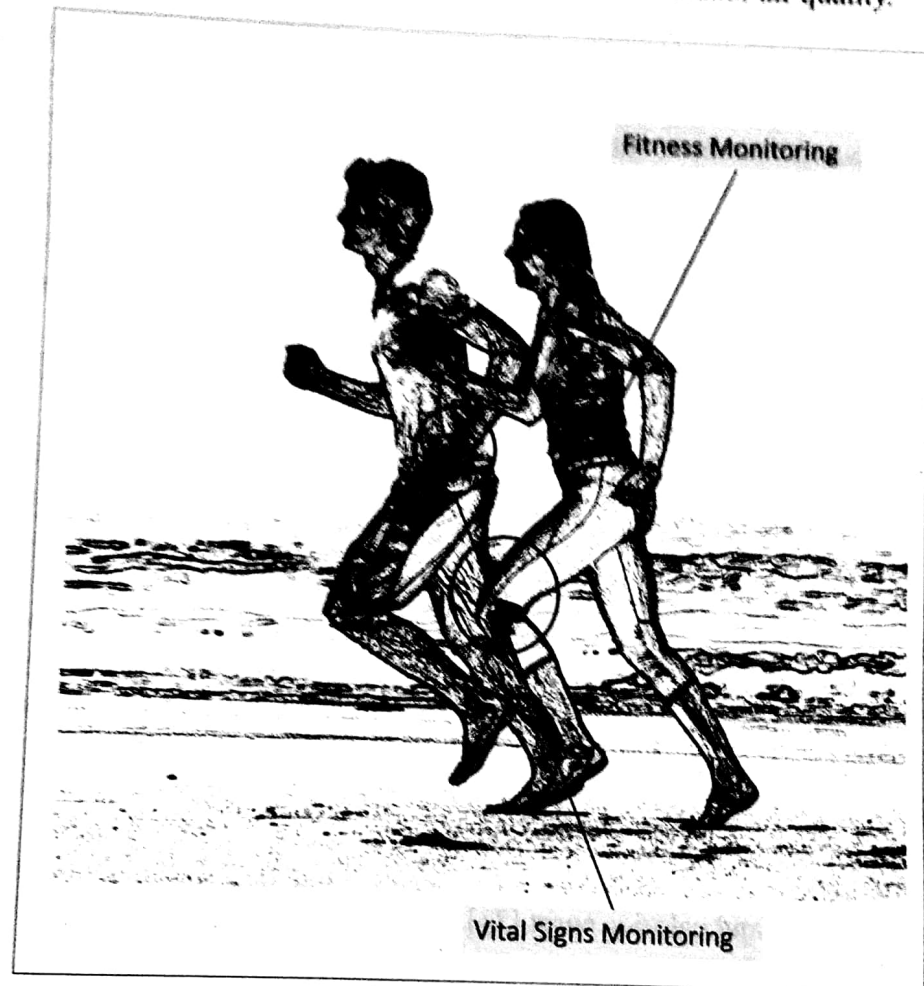


Figure 2.9: Applications of IoT for health

2.10 Health & Lifestyle

2.10.1 Health & Fitness Monitoring

Wearable IoT devices that allow non-invasive and continuous monitoring of physiological parameters can help in continuous health and fitness monitoring. These wearable devices may can be in various forms such as belts and wrist-bands. The wearable devices form a

type of wireless sensor networks called body area networks in which the measurements from a number of wearable devices are continuously sent to a master node (such as a smart-phone) which then sends the data to a server or a cloud-based back-end for analysis and archiving. Health-care providers can analyze the collected health-care data to determine any health conditions or anomalies. Commonly used body sensors include: body temperature, heart rate, pulse oximeter oxygen saturation (SpO₂), blood pressure, electrocardiogram (ECG), movement (with accelerometers), and electroencephalogram (EEG). An ubiquitous mobility approach for body sensor networks in health-care is proposed in [72]. In [73], a wearable ubiquitous health-care monitoring system is presented that uses integrated electrocardiogram (ECG), accelerometer and oxygen saturation (SpO₂) sensors. Fitbit wristband [74] is a wearable device that tracks steps, distance, and calories burned during the day and sleep quality at night.

2.10.2 Wearable Electronics

Wearable electronics such as wearable gadgets (smart watches, smart glasses, wristbands, etc.) and fashion electronics (with electronics integrated in clothing and accessories, (e.g., Google Glass or Moto 360 smart watch) provide various functions and features to assist us in our daily activities and making us lead healthy lifestyles. Smart watches that run mobile operating systems (such as Android) provide enhanced functionality beyond just timekeeping. With smart watches, the users can search the Internet, play audio/video files, make calls (with or without paired mobile phones), play games and use various kinds of mobile applications [68]. Smart glasses allow users to take photos and record videos, get map directions, check flight status, and search the Internet by using voice commands [69]. Smart shoes monitor the walking or running speeds and jumps with the help of embedded sensors and be paired with smart-phones to visualize the data [70]. Smart wristbands can track the daily exercise and calories burnt [71].

Summary

In this chapter you learned about domain specific applications of Internet of Things (IoT). For homes, IoT has several applications such as smart lighting that adapt the lighting to suit the ambient conditions, smart appliances that can be remotely monitored and controlled, intrusion detection systems and smart smoke detectors. For cities, applications of IoT include smart parking systems that provide status updates on available slots, smart lighting that helps in saving energy, smart roads that provide information on driving conditions and structural health monitoring systems. For environment, you learned about IoT applications including weather monitoring, air and noise pollution, forest fire detection and river flood detection

systems. You learned about IoT applications for energy systems including smart grids, grid integration of renewable energy sources and prognostic health management systems. For retail domain, you learned about IoT applications such as inventory management, smart payments and smart vending machines. For agriculture domain, you learned about smart irrigation systems that help in saving water while enhancing productivity and green house control systems. You learned about the industrial applications of IoT including machine diagnosis and prognosis systems that help in predicting faults and determining the cause of faults and indoor air quality systems. You learned about IoT applications for health and lifestyle such as health and fitness monitoring systems and wearable electronics. The applications generate much value to the end users and also provide new revenue opportunities to service and systems providers when integrated to rating, billing and financial applications.

Review Questions

1. Determine the IoT-levels for designing home automation IoT systems including smart lighting and intrusion detection.
2. Determine the IoT-levels for designing structural health monitoring system.
3. Determine the various communication models that can be used for weather monitoring system. Which is a more appropriate model for this system. Describe the pros and cons.
4. Determine the types of data generated by a forest fire detection system? Describe alternative approaches for storing the data. What type of analysis is required for forest fire detection from the data collected?